

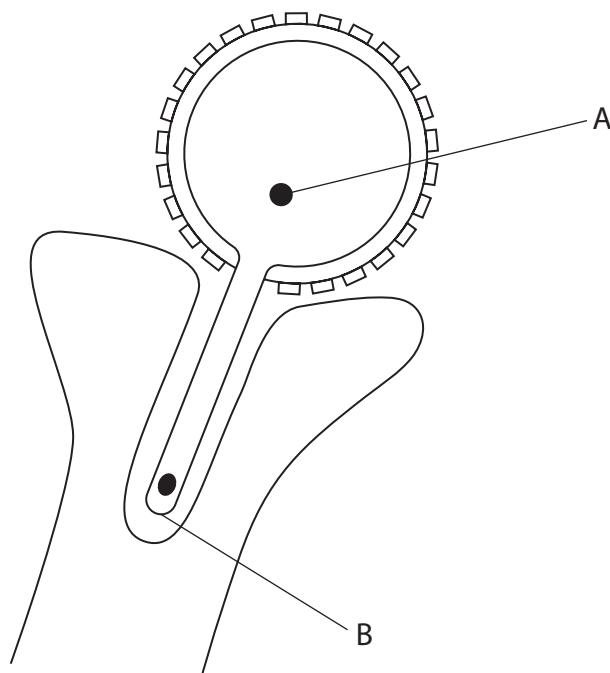
2 Scientists use pollen grains to artificially pollinate certain crop plants.

Collecting pollen by hand is very time consuming.

However, bees collect large amounts of pollen in their daily activity.

Pollen traps can be used to remove pollen when the bees return to their hive.
The viability of the pollen collected by these two methods was investigated.

(a) The diagram shows a germinating pollen grain.



Give a function for each of the structures labelled A and B.

(2)

A

.....

.....

B

.....

.....

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- (b) The percentage germination of the pollen grains was used to determine the viability of the pollen.
Describe a procedure to compare the percentage germination of pollen collected by the two methods used in this investigation.

(5)



- (ii) Describe how the investigation could be modified to allow the significance of the difference between the percentage germination for bee collected pollen and hand collected pollen to be determined at 30 °C.

(4)

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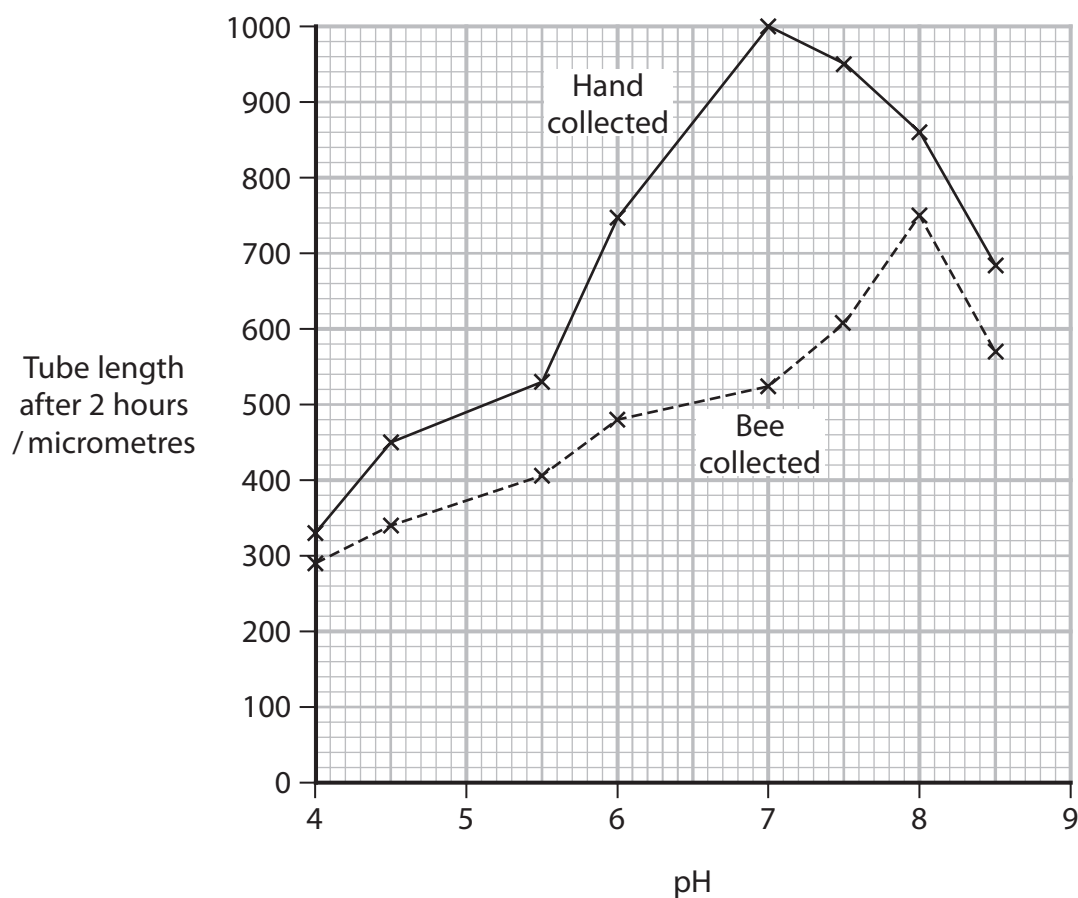
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- (d) The effect of pH on pollen tube growth of pollen collected by these two methods was also investigated.

The results are shown in the graph.



- (i) Calculate the percentage difference in pollen tube length between the optimum pH for hand collected pollen and the optimum pH for bee collected pollen.

Give your answer to three significant figures.

(3)

Answer%

- (ii) Suggest **one** reason for the differences in tube growth of the pollen collected by these two methods.

(1)

(Total for Question 2 = 19 marks)

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